A

Project Report

on

Smart Bus Management System for College Transport Optimization

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “Smart Bus Management System for College Transport Optimization” which is submitted by above students in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE(AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Mr. Shivansh Prasad **Dr. Rekha Kashyap**

Assistant Professor **(Head of Department)**

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ABSTRACT

Efficient transportation is critical for student and faculty mobility in educational institutions. Traditional bus management systems often lack real-time tracking, accurate scheduling, and safety features. This paper presents a Smart Bus Management System (SBMS) designed to address these issues by providing real-time bus tracking, precise stop arrival predictions, and comprehensive user management. Leveraging modern web technologies, GPS integration, the system enhances transparency, reduces waiting times, and improves commuter safety. A case study conducted in a college environment demonstrates the system's feasibility and practical benefits, highlighting its potential to modernize campus transportation management.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Transportation is an essential part of daily life at colleges and universities, directly affecting how students and faculty get around and how punctual they can be. Yet, despite its importance, many institutions still rely on manual or semi-automated bus management systems. These traditional systems are often inefficient and can leave students and staff in the dark about bus locations or schedules. Common issues include a lack of real-time updates, unreliable communication during delays, and difficulty handling unexpected events like traffic jams or vehicle maintenance. The impact is measurable—according to a 2021 report from the Ministry of Road Transport and Highways, about 15% of student delays were caused by inefficiencies in college transportation. Clearly, there's a real need for a smarter, more reliable solution that can modernize how campuses manage transport.

1.2 PROJECT DESCRIPTION

To tackle these challenges, we developed the Smart Bus Management System (SBMS), a solution designed to make college transportation more transparent, efficient, and safe. At the heart of SBMS is a combination of modern technologies, including real-time GPS tracking, cloud-based databases, and intuitive user interfaces. The project took inspiration from commercial ride-hailing services like Uber and Ola, which are praised for their real-time tracking, easy-to-use apps, and personalized dashboards. By adopting

similar features, SBMS aims to provide students and staff with clear, up-to-date information about bus locations, estimated arrival times, and even occupancy levels.

But SBMS is more than just a tracking tool. It shifts the system from a passive, schedule-based model to an active, real-time platform that empowers all users. Students can see exactly when their bus will arrive, avoid unnecessary waiting, and feel less stressed about transportation uncertainties. For the transport department, the system introduces data-driven decision-making and accountability, helping streamline operations and improve overall efficiency.

CHAPTER 2

LITERATURE REVIEW

2.1 Previous Studies in Bus Tracking

Intelligent Transport Systems (ITS) have been a focal point in both academic research and practical applications, particularly concerning the integration of GPS and IoT technologies for public transit monitoring. Several studies have laid the groundwork for understanding the feasibility and impact of such systems

- Studies by researchers like **Kumar et al. (2022)** and **Sharma (2021)** demonstrated that GPS-based bus tracking is technically feasible and can provide valuable location information.
- **Gunathilake & Sharanjah** (2025) proposed a real-time GPS-based bus tracking system in Sri Lanka aimed at improving public transportation efficiency
- **Balani (2023)** introduced a web-based system that highlighted the routes taken by each bus during its journey. This approach aimed to reduce unnecessary waiting times and improve the overall commuting experience by providing real-time updates on bus locations.

2.1 Recent Trends

Modern research in smart transportation is moving beyond simple tracking. Today, the focus is on real-time, responsive systems that can analyze data on the fly, provide predictive insights, and present information in an easy-to-understand format. For example, a study highlighted the use of IoT sensors and GPS technology to enable real-time tracking of buses, providing invaluable information for passengers to access arrival times and route updates through mobile apps or digital displays at stations and stops

CHAPTER 3

System Architecture and Implementation

3.1 System Architecture

The Smart Bus Management System (SBMS) is designed as a full-stack web application, following a modular architecture inspired by microservices. This approach ensures that the system is easy to maintain, scalable for future expansion, and capable of handling multiple users and routes efficiently.

The frontend, which is the interface for students, drivers, and administrators, is built with React.js. It uses Axios to communicate with the backend, React Router for smooth page navigation, and Tailwind CSS to provide a modern, responsive design.

The backend runs on Node.js and Express.js, creating a fast and non-blocking server environment to handle API requests and business logic. User authentication is managed using JSON Web Tokens (JWT), while bcrypt.js ensures that passwords are encrypted and secure.

All application data—including user details, bus routes, and real-time location information—is stored in a MongoDB database, with Mongoose acting as the object data modeling library for easier database interactions.

Live bus tracking is powered by Google Maps API or Mapbox, which renders real-time bus positions on an interactive map. CI/CD pipeline through GitHub Actions, allowing quick and reliable updates. Sensitive configuration variables are managed using dotenv, and Firebase Cloud Messaging is integrated for push notifications.

3.2 Workflow and User Roles

The SBMS workflow is designed for efficiency and clarity:

1. Data Collection: The driver’s mobile device collects GPS coordinates every 5 seconds.

2. Data Transmission: The location data is securely sent to the backend via HTTPS requests.
3. Data Processing: The backend updates the bus location in MongoDB in real time.
4. Data Display: The frontend fetches the updated location via REST API calls and displays the bus position on a Google Map.

The system also defines distinct user roles, each with tailored access

- Students: Can track their assigned bus, view bus details, and check estimated arrival times at stops.
- Drivers: Have an interface to transmit location data and receive real-time route instructions.
- Admins: Can manage buses, configure routes, oversee users, and access a full administrative dashboard.

3.3 System Components and Technologies

To summarize the technical stack clearly, here's a concise table of SBMS components and their corresponding technologies:

Component	Technology
Frontend	React.js, Tailwind CSS, Axios
Backend	Node.js, Express.js
Database	MongoDB, Mongoose
Authentication	JWT, bcrypt.js
Live Tracking	GPS, Google Maps API / Mapbox
Notifications	Firebase Cloud Messaging
CI/CD	GitHub, GitHub Actions

CHAPTER 4

System Features

4.1 Features

The Smart Bus Management System (SBMS) offers a range of features specifically designed to overcome the limitations of traditional transport management systems. Each feature is aimed at providing a seamless, informative, and user-friendly experience for students, drivers, and administrators alike.

- **Live Location Tracking**: At the heart of SBMS is its ability to display buses' real-time locations. GPS data from the driver's device is updated every 5 seconds, ensuring that users can track buses almost instantly on an interactive map.
- **Real-Time ETA Predictions**: By leveraging geolocation data and advanced routing APIs, the system provides accurate estimated times of arrival (ETAs) for each stop along a bus route. This helps students plan their commute efficiently and reduces waiting times at stops.
- **User Dashboards**: SBMS offers customized dashboards for different user roles. Students, drivers, and administrators each get a tailored interface that shows the information most relevant to them, enhancing usability and clarity.
- **Bus Member Management**: The administrative dashboard includes tools to manage bus rosters effectively. Administrators can easily add, remove, or update bus members, keeping records accurate and up-to-date.
- **Security Measures**: Security is a priority. The system uses JWT-based authentication and secure HTTPS API requests to ensure that all data transmission is encrypted, safeguarding against unauthorized access.

- **Notifications**: An optional push notification feature alerts users to key events, such as bus arrivals and departures, improving communication and convenience for all stakeholders.

CHAPTER 5

Conclusion and Future Scope

5.1 Conclusion

The **Smart Bus Management System (SBMS)** has proven to be a practical, scalable, and student-focused solution for improving campus transportation. Built on a robust cloud-based architecture and inspired by commercial ride-sharing platforms, SBMS effectively addresses the limitations of traditional transport systems.

A pilot deployment across multiple campus routes demonstrated tangible improvements: students could track buses in real time, plan their commutes more efficiently, and spend less time waiting at stops. Administrative staff also benefited from a noticeable reduction in routine inquiries and manual tracking tasks. These outcomes show that SBMS not only streamlines operations but also enhances the overall user experience, creating a smoother, safer, and more reliable transportation system for the entire campus community.

Overall, the project validates the impact that a well-designed, technology-driven solution can have on operational efficiency, user satisfaction, and institutional service quality.

5.2 Future Scope

While the current system lays a strong foundation, there are several opportunities to enhance SBMS further:

- **Predictive Analytics:** By integrating machine learning models that analyze historical traffic and real-time data, the system could provide more accurate ETAs even during heavy traffic, making commuting more reliable.
- **ML-Driven Occupancy and Demand Prediction:** Future versions could predict bus occupancy and student demand at different times of day, allowing for smarter route planning and dynamic bus allocation..
- **Biometric Attendance Integration:** Integrating biometric systems would allow automated tracking of bus ridership, enhancing both security and administrative oversight.

These enhancements directly address current limitations and aim to make SBMS more **resilient, intelligent, and seamlessly integrated** into the campus ecosystem. With these upgrades, the system could evolve into a comprehensive platform that not only tracks buses but actively optimizes transportation and improves the daily commuting experience for students and staff.

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